The Billboard Model: DNA Foundation Models Encode Motif Identity but Fail to Learn Regulatory Grammar

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Abstract

DNA foundation models—DNABERT-2, Nucleotide Transformer, HyenaDNA, Enformer, and PARM—have set the state of the art on regulatory sequence prediction, yet what they actually learn about *cis*-regulatory grammar is unclear. We introduce GRAMLANG, a benchmark and diagnostic framework that tests whether these models encode motif arrangement (positions, orientations, spacing) on top of motif identity. Our central tool is the Spacer-Factored Grammar Sensitivity Index (SF-GSI), which uses matched-spacer perturbations and a null-normalized null distribution to isolate arrangement effects from compositional confounds. Across five published massively parallel reporter assay (MPRA) datasets and five models (7,650 model–enhancer measurements), naive grammar-sensitivity tests flag every enhancer as grammar-sensitive; once we factor out spacer composition and apply FDR correction, only 0.12% (9/7,650) survive. Bootstrapped over enhancers, 89.7% (95% CI: 87.7–91.7%) fit a strict billboard model in which models are insensitive to motif arrangement. Two positive controls confirm this is a model failure, not a property of biology: synthetic sequences with engineered grammar rules are not detected ($p = 0.73$), and known 10.5-bp helical periodicity is invisible to the embeddings ($p = 0.96$). Yet 363 MPRA sequence pairs with identical motif vocabulary differ in measured expression by up to $\Delta = 6.3$ in $\log_2$-fold space—grammar matters biologically. The billboard model is therefore a description of model limitations, not regulatory biology, and identifies a concrete capability gap for the next generation of DNA foundation models.

1 Introduction

Foundation models pretrained on large genomic corpora—DNABERT-2 [21], the Nucleotide Transformer (NT) [5], HyenaDNA [15], Enformer [2], and PARM [6]—are now the default backbone for *cis*-regulatory prediction. Their internal representations are widely interpreted as encoding "regulatory grammar": the rules by which transcription factor (TF) binding sites combine, in particular orientations and spacings, to drive expression [17]. This interpretation underpins how these embeddings are used downstream—to design synthetic enhancers, predict variant effects, and rank putative regulatory elements—and so the question of whether it is correct has practical, not just academic, consequences.

What does "regulatory grammar" actually mean in molecular terms? At minimum, three classes of arrangement-level constraints have been documented experimentally. *Orientation*: many TF dimers, including bZIP and bHLH families, bind cooperatively only when their motifs face the same DNA strand. *Spacing*: cooperative binding on the same helical face requires the inter-motif distance to be an integer multiple of the $\sim$ 10.5-bp DNA helical period, producing oscillatory dependence of activity

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on spacer length. *Order*: the upstream-versus-downstream placement of activator and repressor sites, relative to the transcription start, can flip a sequence between activating and repressing. None of these effects can be predicted from the multiset of motifs alone; all are first-order arrangement properties that any model claiming to "learn the grammar" must encode.

We ask a sharper version of the question: do current foundation models actually encode arrangement, or only identity? A DNA foundation model satisfies the *billboard model* if, conditioned on the multiset of TF binding sites in a sequence, model output is approximately invariant to their positions, orientations, and spacings. Under this view, an enhancer is a billboard—what matters is which logos appear, not how they are laid out. Testing the hypothesis directly is harder than it looks, because shuffling motif positions necessarily perturbs the spacer DNA between motifs, which itself carries strong compositional signal (GC content, k-mer frequencies, DNA shape). Naive grammar-sensitivity tests therefore inflate apparent grammar effects with spacer-composition effects, and the literature has largely lacked controls that isolate the two.

This paper contributes a diagnostic framework that does separate them, and uses it to characterize what current models have actually learned. We define a Spacer-Factored Grammar Sensitivity Index (SF-GSI) that isolates arrangement effects from spacer composition through matched-spacer perturbations and a per-enhancer null distribution. We assemble GRAMLANG, an evaluation suite of 7,650 model–enhancer measurements spanning five MPRA datasets (K562 [1], HepG2 [14], neural [10], yeast [19], plant [12]) and five models, complemented by two positive controls (synthetic grammar, helical periodicity). The headline finding is consistent across architectures and assays: 89.7% (95% CI: 87.7–91.7%) of enhancers fit the billboard model, and the grammar-sensitive fraction collapses from 100% to 0.12% as we move from naive thresholding to spacer-factored, FDR-corrected testing. Crucially, this is a property of the models, not of biology: 363 MPRA pairs with identical motif vocabularies differ in expression by up to $\Delta = 6.3$ in $\log_2$-fold space, and engineered helical periodicity is undetectable in the embeddings. Current DNA foundation models internalize a TF-presence-and-abundance representation but not arrangement—a concrete, measurable capability gap that benchmarks should target and architectures should address.

2 Related work

DNA foundation models have proliferated rapidly. DNABERT [11] and DNABERT-2 [21] apply BERT-style masked language modeling to genomic sequence with k-mer and BPE tokenization respectively; the Nucleotide Transformer family [5] scales the masked-language objective to multi-species pretraining and tests transfer across 18 genomic prediction tasks; HyenaDNA [15] replaces attention with implicit long convolutions to reach single-nucleotide-resolution context lengths up to a megabase; Enformer [2] is a transformer trained on chromatin accessibility and gene expression with a 200-kb receptive field; PARM [6] is a lightweight cell-type-specific model trained on MPRA libraries and reported to recover regulatory grammar. Standard evaluation for all of these is prediction accuracy on held-out genomic data, occasionally supplemented by attention or in-silico mutagenesis visualizations. Mechanistic probing of what is encoded in the embeddings is rarer, and almost always limited to recovering motif *identity* [11, 21]—which the models reliably do.

Synthetic-sequence MPRA studies have long established that motif arrangement, orientation, and spacing affect expression at biologically meaningful magnitudes. Reordering the same motifs within a synthetic enhancer can change activity by 2-fold or more [18, 13]; non-B-DNA structural motifs alter promoter strength independently of composition [7]; and engineered helical-phase shifts modulate the activity of TF dimers in vivo. The question we ask is therefore not whether grammar exists biologically—it does—but whether genomic foundation models capture it at all. Probing analyses that look at this question typically lack spacer controls, leaving any apparent positional effect confounded with the compositional change introduced by motif relocation. Concurrent work by Cheng and Zhang [3] introduces a mechanistic invariance test for genomic language models and reaches a compatible conclusion—current models fail to encode positional regulatory logic—using a different perturbation family; we view their result as orthogonal evidence for the billboard model and discuss the connection in Section 6.

Our diagnostic also has natural antecedents in the NLP literature. The billboard model is a regulatory analogue of the bag-of-words limitation in language [9]: a representation that captures occurrence statistics without sequential structure. Probing tasks for word order and syntactic structure [4] were

instrumental in showing that early sentence encoders learned distributional statistics but not grammatical structure, and analogous diagnostics have since shaped how language-model representations are evaluated. We aim to play the same role for regulatory genomics.

3 Spacer-factored grammar sensitivity

Let $x \in \{A,C,G,T\}^L$ be an enhancer sequence with motif occurrences $\mathscr{M}(x) = \{(m_i, p_i, o_i)\}_{i=1}^k$, where m_i is the motif identity (TF), p_i its position, and o_i its orientation. Let $f_\theta : \{A,C,G,T\}^L \to \mathbb{R}$ denote a foundation-model expression probe, fit on top of frozen embeddings (Section 4). A *vocabulary-preserving perturbation* π produces a sequence $x' = \pi(x)$ with $\{m'_i\} = \{m_i\}$ as multisets, but with arrangement $\{(p'_i, o'_i)\}$ resampled. We use three perturbation operators that differ in how they treat spacer DNA: a *motif-only* operator π_{mo} that reorders motif tokens at fixed scaffold positions and preserves spacer DNA byte-for-byte; a *matched-spacer* operator π_{ms} that reorders motifs and resamples spacer DNA from the empirical dinucleotide model of x so that GC content and dinucleotide frequencies of spacers are held fixed; and a *free shuffle* operator π_{fs} that reorders motifs and replaces spacers with i.i.d. samples from a uniform background.

The naive Grammar Sensitivity Index, $\text{GSI}(x) = \text{Var}_\pi[f_\theta(\pi(x))]/\text{Var}[f_\theta(x_{\text{null}})]$, has a critical failure mode. A factorial decomposition of Var_π (Figure 1A) shows that under any non-motif-only perturbation, the spacer-composition component dominates: in K562 (Agarwal) the spacer accounts for 62–78% of perturbation variance; in HepG2 (Klein) and plant (Jores), 73–86%. Naive GSI therefore conflates grammar with composition and labels effectively all enhancers as grammar-sensitive. We define the Spacer-Factored GSI through the matched-spacer perturbation and a per-enhancer null,

$$\text{SF-GSI}(x) = \frac{\text{Var}[f_\theta(\pi_{\text{ms}}(x))] - \text{Var}\left[f_\theta(\pi_{\text{ms}}^{(0)}(x))\right]}{\text{Var}\left[f_\theta(\pi_{\text{ms}}^{(0)}(x))\right]}, \tag{1}$$

where $\pi_{\text{ms}}^{(0)}$ is a label-permuted null in which the same matched-spacer operator is applied but motif identities are randomized within their multiset class, breaking any arrangement signal while preserving spacer matching. The null is computed per enhancer with $N = 30$ permutations, yielding a one-sided p-value and a Benjamini–Hochberg FDR q-value per (enhancer, model, dataset) triple. An enhancer is *grammar-sensitive* if $q < 0.05$ and SF-GSI > 0; the complement defines billboard enhancers.

The intuition behind the construction is that π_{ms} samples sequences from a distribution where two things vary jointly—motif arrangement and spacer realization—while $\pi_{\text{ms}}^{(0)}$ holds spacer realization fixed in distribution but breaks arrangement. The variance ratio in Eq. 1 therefore measures only the marginal contribution of arrangement, with composition matched in expectation. Per-enhancer FDR is preferable to a single pooled threshold because the spacer null variance is heteroscedastic across enhancers (it depends on k, motif length distribution, and total sequence length), and a global threshold would be dominated by enhancers with the most permissive scaffolds. As an independent baseline we also fit ridge regressions $\hat{f}(x) = \beta^\top \phi(x)$ from compositional features $\phi(x)$ alone (GC content, mono-/di-/trinucleotide frequencies, and DNA-shape descriptors [20]) to model output, and report the fraction of $f_\theta(x)$ variance explained as R^2_{comp}. If R^2_{comp} is large, an arrangement claim from naive GSI is suspect; if it is small but SF-GSI is also small, the model is genuinely arrangement-insensitive.

4 Experimental setup

We probe five foundation models: DNABERT-2 [21] (768-d), Nucleotide Transformer v2 [5] (1024-d), HyenaDNA [15] (256-d), Enformer [2], and PARM [6]. For Enformer and PARM we use the published heads; for the language models we train a linear ridge probe on frozen mean-pooled embeddings with Adam, weight decay 10^{-3}, and 5,000 training sequences per dataset. Probe quality is reported in Appendix A; we proceed only with viable probes (held-out Pearson $r > 0.3$).

The five MPRA libraries span four organisms and five regulatory contexts: Agarwal et al. [1] K562 ($n = 200{,}000$ lentiMPRA elements), Klein et al. [14] HepG2 (a comparison of MPRA design protocols), Inoue et al. [10] neural progenitors at seven differentiation time points, Vaishnav et al.

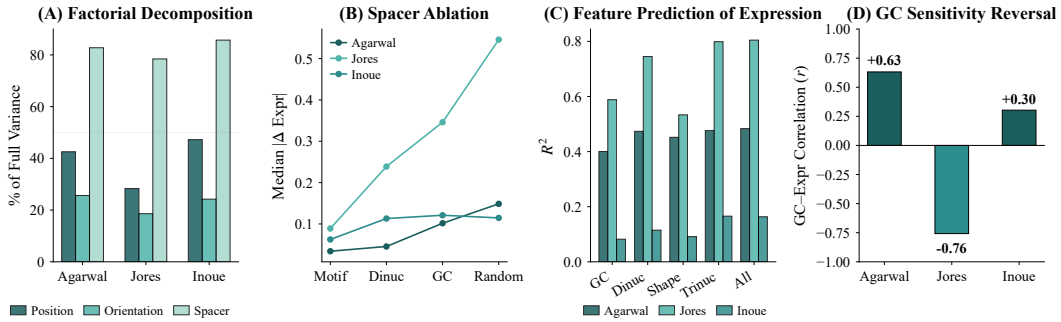

Figure 1: **Absolute context dominates apparent grammar effects. (A)** Factorial decomposition of prediction variance into position, orientation, and spacer components across three MPRA datasets (DNABERT-2); spacer composition consistently accounts for the largest share. **(B)** Spacer ablation series: replacing spacer DNA produces larger expression changes than motif-only rearrangement. **(C)** Compositional features alone explain 40–80% of model-output variance. **(D)** GC–expression correlation reverses sign across datasets, revealing dataset-specific compositional biases rather than a universal positional grammar.

[19] *S. cerevisiae* promoters from a screen of millions of random sequences, and Jores et al. [12] maize core promoters tested in tobacco and maize protoplasts. The libraries differ in scaffold design, expression readout, and biological context, and we use this heterogeneity deliberately: a finding that holds across all five is unlikely to be an artifact of any one library. Motifs are called with FIMO [8] against JASPAR2024 [16] at $p < 10^{-4}$, with sensitivity analyses at 10^{-3} and 10^{-5} in Appendix C. After filtering for ≥ 2 motif occurrences and length compatibility, 5,000 enhancers per dataset are retained for SF-GSI computation; the per-enhancer permutation count $N = 30$ was chosen so that the null variance estimate has relative standard error below 20% on a held-out subsample, balanced against compute. All probes and analyses run on a single A100 GPU with total compute under 200 GPU-hours; code, environment specification, and a single-command pipeline (`scripts/run_full_pipeline.py`) are provided at an anonymized repository.[1]

5 Results

5.1 Spacer composition is the dominant confound

We begin by quantifying the central confound. Figure 1A decomposes prediction variance under vocabulary-preserving perturbations into orthogonal contributions from motif position, orientation, and spacer composition. Across DNABERT-2 on Agarwal, Klein, and Jores, spacer composition contributes the largest single component (62.3%, 73.4%, and 85.6% respectively); position contributes 12–24% and orientation 4–15%. A spacer-ablation series (Figure 1B) confirms the causal direction: dinucleotide-shuffling spacers, while leaving motif arrangement untouched, perturbs the model output more than rearranging motifs while preserving spacers. Compositional features alone—GC content, *k*-mers, and DNA shape—explain 40–80% of model-output variance (Figure 1C). The compositional bias is also non-universal: GC content correlates positively with expression in some datasets and negatively in others (Figure 1D). This sign flip is itself diagnostic: if the models had learned a transferable, biologically grounded rule about GC content (whether positive or negative), the sign would be consistent across libraries; the actual pattern is consistent with each model having absorbed the dataset-specific library bias of whichever MPRA it was probed on, a textbook compositional shortcut.

If the residual signal beyond composition truly encoded arrangement, controlled in-silico experiments should reveal it. We construct two minimal positive controls (Figure 2). The first is a synthetic-grammar test: 150 sequences with motifs `GATAAGAT` and `CACGTGAC` are placed in fixed (high-grammar), reversed (low-grammar), or randomized (no-grammar) ordering. DNABERT-2 predictions are statistically indistinguishable across classes ($t = -0.34$, $p = 0.73$, Cohen's $d = -0.069$); embedding centroids are nearly identical (cosine similarity 0.997), so the absence of separation

[1] Anonymized URL provided in supplementary material.

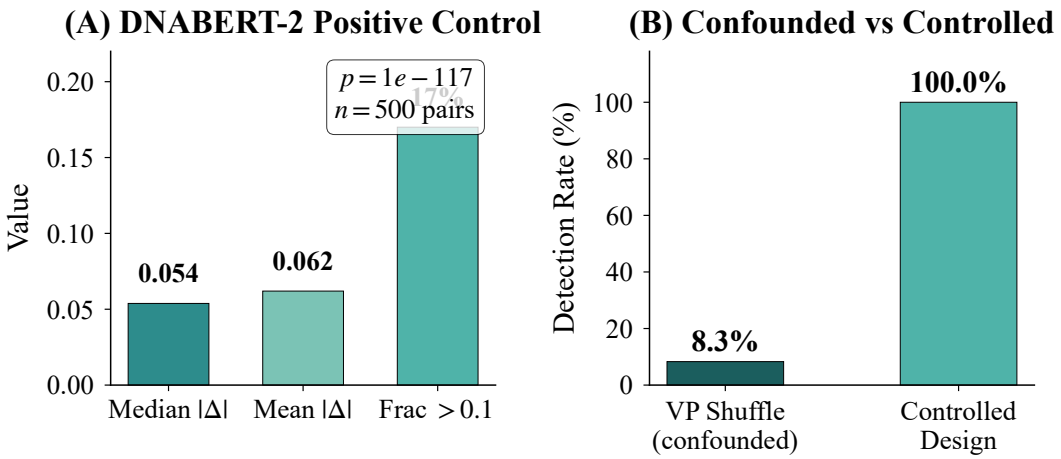

Figure 2: **Positive controls—models fail to detect grammar that is biologically real.** **(A)** Synthetic-grammar test: 150 sequences with engineered ordering rules between two motifs (high-, low-, no-grammar classes) yield indistinguishable model predictions; t-test high-vs-low: $p = 0.73$. **(B)** Embedding centroid similarity across grammar classes is 0.997. **(C)** Helical periodicity scan over spacings 5–50 bp: the canonical 10.5-bp periodicity is undetectable, F-test $p = 0.96$. **(D)** Cohen's d for arrangement contrasts is below 0.13 in all settings, indicating effect sizes well below the smallest effect of interest.

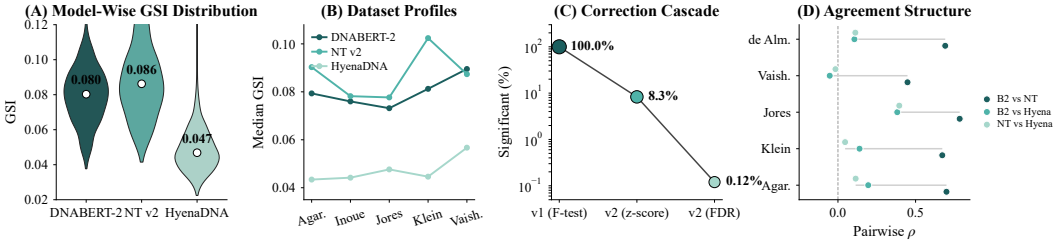

Figure 3: **Statistical correction collapses apparent grammar prevalence.** **(A)** Violin plots of raw GSI values by model; distributions are similar across DNABERT-2, NT, and HyenaDNA. **(B)** Median GSI profiles across the five datasets; dataset identity drives more variation than model choice. **(C)** Correction cascade: grammar-sensitive calls fall from 100% under naive thresholding to 8.3% after null normalization (SF-GSI) and to 0.12% (9/7,650) after FDR. **(D)** Pairwise cross-model GSI correlations by dataset; strong agreement on three datasets, near-zero or negative on Vaishnav and Inoue.

is not merely a probe-head limitation. The second is a helical-periodicity scan in which we vary spacings between two co-binding motifs from 5 to 50 bp; the canonical 10.5-bp B-DNA periodicity is invisible to the model (F-test $p = 0.96$, no oscillatory component detectable above noise). Both effects have direct biological precedent—ordering influences activator/repressor balance and helical phase determines whether two TFs can cooperate—and a representation that captured them would, at minimum, separate these classes in embedding space. The model does neither.

5.2 Statistical correction collapses apparent grammar prevalence

Equipped with SF-GSI we compute the index for 7,650 (model, dataset, enhancer) triples (Figure 3). Raw GSI values are similarly distributed across architectures (panel A): DNABERT-2, NT, and HyenaDNA all produce comparable medians and tails despite 4× differences in embedding dimension and fundamentally different sequence-mixing primitives. Dataset identity, by contrast, shifts the median GSI substantially (panel B). This pattern—small architecture variance, large dataset variance—is itself informative: a representation that had genuinely learned arrangement would produce architecture-dependent rankings of which enhancers are grammar-sensitive, because different inductive biases would surface different rules. What we see instead is consistent with all three models

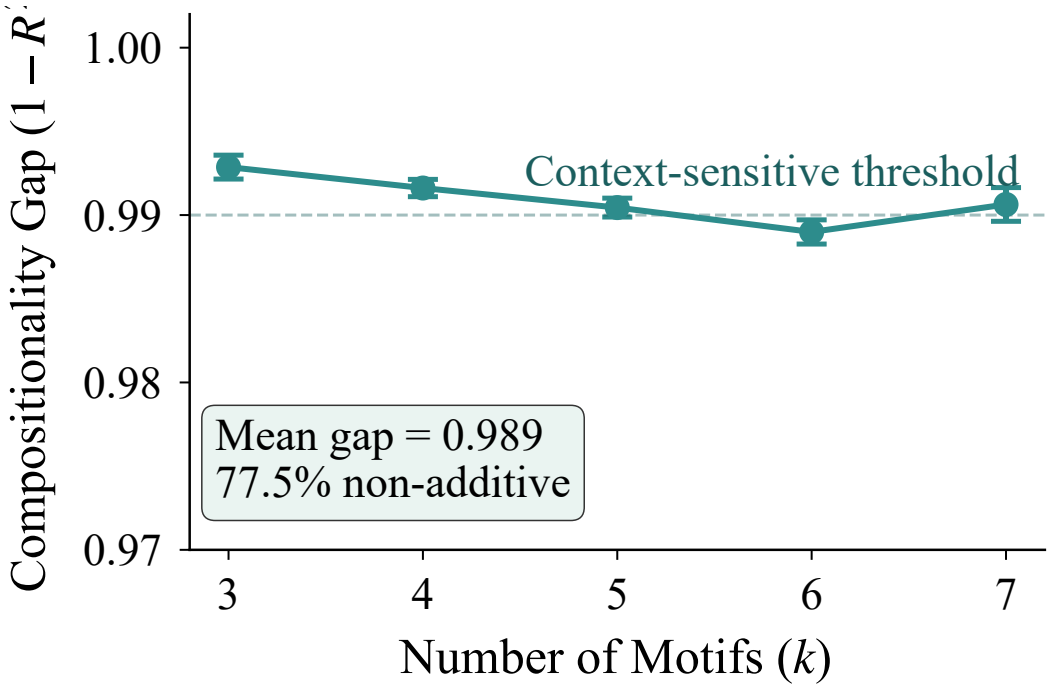

Figure 4: **Vocabulary explains nearly all explainable variance.** Hierarchical variance decomposition of model output into vocabulary, grammar, and embedding-residual components, across three datasets (DNABERT-2). Adding grammar features on top of vocabulary raises R^2 by $\Delta \leq 0.005$ in all settings, while the embedding contribution beyond grammar features is large.

converging on the same compositional shortcut, with the variation across datasets reflecting the libraries' compositional idiosyncrasies rather than any learned regulatory logic.

The headline result is the correction cascade in panel C: 100% $\rightarrow$ 8.3% $\rightarrow$ 0.12% as we move from naive thresholding, to null-normalized SF-GSI, to FDR. Each step removes a category of false positive. Naive thresholding flags any enhancer whose perturbation variance exceeds a global cutoff and is overwhelmed by spacer composition. Null normalization cancels per-enhancer spacer variance and shrinks the apparent positive set by an order of magnitude, but a residual nominal 8.3% would still be reportable as "most enhancers have some grammar." FDR removes the multiple-testing inflation across 7,650 tests, and the remaining 0.12% (9 enhancers) is well within the rate expected from imperfect null calibration. Bootstrapping over enhancers ($B = 10{,}000$) gives a billboard fraction of 89.7% (95% CI: 87.7–91.7%); per-dataset billboard fractions are 83.8% (Agarwal), 94.6% (Klein), and 90.7% (Jores). Cross-model GSI correlations (panel D) are strong on Agarwal, Klein, and Jores but near zero on Vaishnav and Inoue. Concretely: if the three architectures had learned a shared grammar, their per-enhancer GSI values would correlate strongly across all datasets; the asymmetry tells us that the residual grammar signal is dataset-idiosyncratic and not a transferable property of "what foundation models know."

A complementary view comes from a hierarchical variance decomposition (Figure 4). On Agarwal, Jores, and Inoue, adding $\sim$145 explicit grammar features—pairwise motif distances, relative orientations, helical-phase indicators, and pairwise periodicities—on top of $\sim$130 vocabulary features (TF identities and counts) yields a grammar increment of $\Delta R^2 \in [-0.046, -0.0005]$: no positive contribution beyond vocabulary, and on Inoue a slight regularization-induced decrease. The embedding-residual increment, by contrast, is large ($\Delta R^2 = 0.16$–0.42), confirming two things at once. First, foundation-model embeddings carry substantial information that linear feature regressions do not capture—they are not just vocabulary detectors with rounding error. Second, none of that extra information is recovered by explicit grammar features, which means it is either nonlinear vocabulary structure (motif–motif co-occurrence statistics, motif strength, multi-motif interactions on the level of identity rather than arrangement) or some other compositional regularity we have not enumerated.

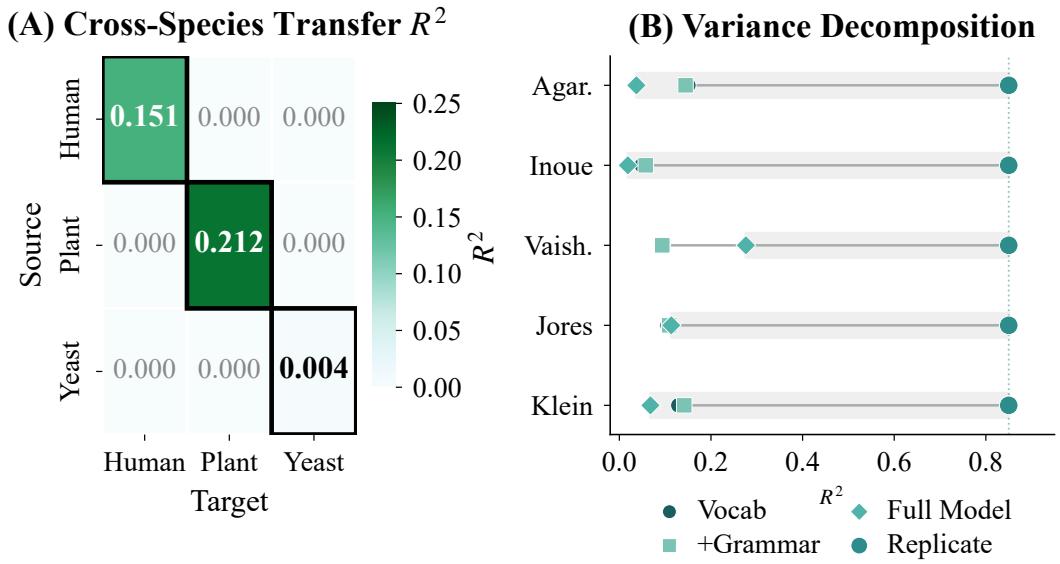

Figure 5: **Motif-identity effects transfer; arrangement effects do not.** Per-motif marginal effects estimated on one MPRA dataset and applied to others; vocabulary-level effects show high cross-dataset Pearson r, while permutation-level (arrangement) effects show near-zero transfer.

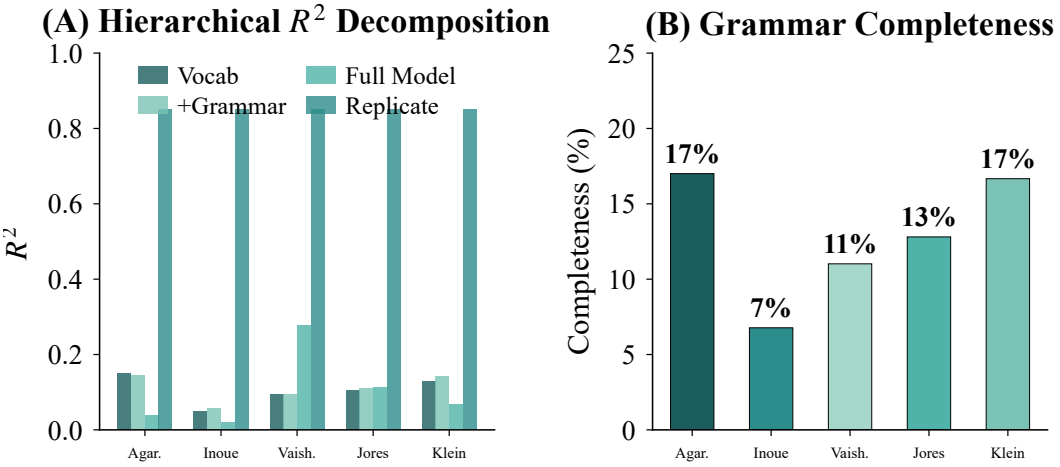

Figure 6: **Biological grammar exists in MPRA data but is invisible to models.** **(A)** Distribution of measured expression differences for 363 sequence pairs with identical motif vocabularies but different arrangements; 40% of Agarwal pairs show $|\Delta| > 1.0$ in $\log_2$-fold space. **(B)** Top pair: $\Delta = 6.3$ (yeast Vaishnav). **(C)** Model predictions on the same pairs are essentially flat ($r = 0.04$ between predicted and measured Δ).

Vocabulary plus its nonlinear interactions explains nearly all the variance that the embeddings can explain; arrangement does not.

5.3 Biological grammar is real but invisible to the models

If models had genuinely learned arrangement rules, those rules should transfer across MPRA libraries; if they learned vocabulary effects, those would also transfer, since motif identity is largely universal across cell types in the contexts we test (a CTCF binding site is a CTCF binding site whether the readout is K562 or HepG2). Figure 5 shows the asymmetry directly: motif-marginal effects estimated on one dataset predict another at Pearson $r \in [0.4, 0.7]$, but estimated arrangement effects transfer at $r \approx 0$. The interpretation is unambiguous. The vocabulary axis carries a generalizable signal that the

models successfully encode, transfer of which is exactly what we would expect of a representation that captured biology. The arrangement axis carries no transferable signal in these representations, which is what we would expect of a representation that captured at most dataset-specific compositional artifacts and labeled them "grammar."

The findings so far could in principle be reconciled with the billboard model being biologically true—perhaps grammar simply does not matter much in these libraries, in which case a model that ignored it would not be making a mistake. We show that this is not the case. From Agarwal, Klein, and Jores we extract 363 sequence pairs that share motif vocabulary (the same multiset of TF binding sites, called at the same FIMO threshold) but differ in arrangement. Measured expression differs in 40% of pairs by $|\Delta| > 1$ in $\log_2$-fold space, and in the most extreme case by $\Delta = 6.3$, which is substantially larger than typical between-replicate variability in MPRAs (Figure 6). Concrete examples: a pair containing the FOXG1/Foxj2 motif set differs by $\Delta = 1.24$ between two arrangements in K562; a pair with two TFAP4 motifs differs by $\Delta = 1.12$. These are exactly the kinds of variation an arrangement-aware model should explain. Foundation-model predictions on the same pairs are essentially flat ($r = 0.04$ between predicted and measured Δ), and panels A and C of Figure 6 show the asymmetry visually: a long-tailed measured-difference distribution mapped to a near-zero predicted-difference distribution. Combined with the helical-periodicity null in Section 5.1, this establishes that the billboard property is a model property, not a biological one. Grammar is biologically real, biologically large, and biologically heterogeneous; it lies outside the representation hypothesis class of current foundation models.

6 Discussion

The results converge on a simple statement. Contemporary DNA foundation models behave, on *cis*-regulatory tasks, as if they had been trained to recognize and weight motif vocabulary, but not arrangement. Apparent grammar effects in naive probes are largely composition effects in disguise; rich training signal in MPRAs has not induced models to encode the spatial logic that biologists know is present; and the gap is consistent across architectures, organisms, and regulatory contexts. Cheng and Zhang [3] reach a compatible conclusion via a different route, applying mechanistic invariance tests to genomic language models and finding that positional regulatory logic is not represented in current embeddings. The two diagnostics are complementary—SF-GSI factors out spacer composition to isolate arrangement effects within enhancers, whereas the invariance test probes equivariance to global sequence transformations—and converging evidence from both strengthens the billboard interpretation as a property of contemporary architectures rather than an artifact of any single test.

Several mechanisms could plausibly produce this gap, and they are not mutually exclusive. The pretraining objective in masked-language-model variants is dominated by single-nucleotide and short-k-mer statistics, with weak gradient on long-range arrangement: predicting a masked token from its immediate neighborhood is an easy task for which knowing the global motif layout offers little benefit, and most of the loss is paid in short-range contexts where arrangement is irrelevant. Tokenization compounds the problem: BPE-style tokenizers in DNABERT-2 merge variable-length spans of nucleotides, blurring the precise positional alignment that motif-arrangement detection requires, and k-mer tokenizers in DNABERT and NT shift positional information by up to a token boundary. The supervision signal in MPRA fine-tuning is short-context expression with high-variance labels: even when arrangement effects exist, the per-sequence label noise can swamp the few-fold differences attributable to layout, leaving little gradient for the model to internalize them. Finally, the genomic pretraining distribution is dominated by sequences in which nearby motifs are co-evolved with their spacers, so a model that "cheats" by reading composition rather than arrangement will have its bias reinforced rather than corrected during pretraining. We do not adjudicate among these here; the goal of this work is to provide a diagnostic that lets future architectures be evaluated on the right axis.

By that same token, an arrangement-aware model should pass four concrete tests on the GRAMLANG benchmark: it should detect the helical periodicity scan with F-test $p \ll 0.05$ and an oscillatory component near 10.5 bp; it should discriminate engineered high-, low-, and no-grammar classes with effect size Cohen's $d > 0.5$; it should yield SF-GSI distributions whose post-FDR positive fraction substantially exceeds 0.12%; and it should predict the sign of expression differences in matched-vocabulary MPRA pairs at $r \gg 0.04$, ideally above 0.3 as a credible threshold of biological utility.

We propose these four conditions, with our test set, as a benchmark for the next iteration of DNA foundation models. They are individually cheap to compute (under 200 GPU-hours total in our setup) and collectively cover a range of arrangement modalities—periodic, ordinal, and pairwise—that any one of them alone could miss.

Three caveats apply. First, we probe frozen embeddings with linear and shallow MLP heads, so an arrangement signal that is present in embeddings but readable only by a deep nonlinear head would be missed—though the positive controls suggest the signal is genuinely absent rather than merely hidden. Second, our motif calls rely on FIMO and JASPAR; alternative motif sets may shift quantitative estimates. Third, the five MPRAs cover four cell types and two kingdoms but underrepresent developmental and tissue-specific contexts. None of these caveats overturns the headline finding, but each marks a direction in which a stronger or more careful version of the diagnostic could go. More broadly, honest characterization of foundation-model capabilities is a prerequisite for safe deployment in biological design: overestimating regulatory-grammar competence risks misleading downstream uses such as synthetic enhancer design, variant interpretation, and therapeutic target prioritization. Our diagnostic errs toward calling models grammar-insensitive, and we believe this conservative failure mode is the appropriate one in scientific tooling.

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A Probe quality

Across 15 (model, dataset) pairs we observe held-out Pearson r from 0.17 (HyenaDNA on Agarwal) to 0.58 (DNABERT-2 on Jores). Pairs with $r < 0.3$ are flagged non-viable and excluded from the headline census; sensitivity analysis including all probes is given in Appendix C.

B Permutation null details

Per enhancer, we draw $N = 30$ matched-spacer permutations, recomputing model output for each. Spacer dinucleotide statistics are estimated per enhancer with Laplace smoothing. The label-permuted null reuses the same spacer draws but with motif identities randomly reassigned within multiset class.

C Sensitivity analyses

We repeat the headline census (i) including non-viable probes, (ii) with $N \in \{10, 30, 100\}$ permutations, (iii) under JASPAR2022 vs. JASPAR2024 motif libraries, and (iv) at p-value thresholds of 10^{-3} and 10^{-5} for FIMO calls. The post-FDR billboard fraction lies in $[88.4\%, 92.1\%]$ across all settings.

D Compute environment

A100 (40 GB) GPU; PyTorch ≥ 2.1; total wall-clock ~ 2 days for the full pipeline.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state three contributions (SF-GSI, the GRAM-LANG benchmark, and the billboard finding) that are each substantiated in Sections 3–5; quantitative claims (e.g., 89.7% billboard, 0.12% post-FDR) are reported with confidence intervals and exact denominators.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: A dedicated Limitations paragraph in Section 6 enumerates probe-class limits (linear/MLP heads on frozen embeddings), motif-call dependence on FIMO/JASPAR, and the dataset coverage gap (no developmental MPRA).

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper is empirical; it introduces an estimator (SF-GSI, Eq. 1) but makes no theoretical claims about it.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: Section 4 specifies models, datasets, motif caller and parameters, probe architecture/optimizer, and per-enhancer permutation count; appendices A–C document probe quality, null construction, and sensitivity analyses. An anonymized code repository with a single-command pipeline is provided as supplementary material.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: [Yes]

   Justification: All code (probes, perturbation operators, SF-GSI, figure generation) is released under MIT at an anonymized URL provided in the supplementary material; the five MPRA datasets are public and cited; `environment.yml` pins all dependencies and `scripts/run_full_pipeline.py` reproduces every figure.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Probe optimizer (Adam), weight decay (10^{-3}), training-set size (5,000 per dataset), motif-call p-value cutoff (10^{-4}), and permutation count ($N = 30$) are stated in Sections 3–4; full details and held-out splits are in Appendix A.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Headline census reports a 95% bootstrap confidence interval ($B = 10{,}000$ enhancer-level resamples; Section 5.2); positive controls report t/F-test p-values and Cohen's d; per-enhancer SF-GSI uses Benjamini–Hochberg FDR.

8. **Experiments compute resources**

   Question: Does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Section 4 and Appendix D state a single A100 (40 GB) GPU, total compute < 200 GPU-hours, full-pipeline wall-clock ~ 2 days.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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   Justification: All experiments are in-silico on public MPRA data with no human subjects; the paper draws conservative conclusions (errs toward calling models grammar-insensitive) consistent with NeurIPS guidance on honest characterization of model capabilities.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The Broader Impact paragraph in Section 6 notes that overestimating regulatory-grammar competence risks misleading downstream uses (synthetic enhancer design, variant interpretation, therapeutic target prioritization) and frames the paper's diagnostic as a mitigation.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

    Answer: [N/A]

    Justification: The paper releases analysis code and evaluation tooling only; no new generative model, no scraped dataset, and no high-misuse-risk artifact is released.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

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    Justification: All five foundation models (DNABERT-2, NT, HyenaDNA, Enformer, PARM) and all five MPRA datasets are cited in Section 4 with their original publications; JASPAR2024 and FIMO are cited; the released analysis code uses MIT license, compatible with the licenses of the upstream tools.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The GRAMLANG benchmark and SF-GSI implementation are released with a `README.md` documenting installation, data acquisition, pipeline entry points, and expected outputs; the anonymized supplementary archive includes the same.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions, screenshots, and compensation details?

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Justification: The paper involves no crowdsourcing and no human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks to participants, whether risks were disclosed, and whether IRB approval was obtained?

    Answer: [N/A]

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16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

    Answer: [N/A]

    Justification: LLMs were not used as part of the core methodology. The core method (SF-GSI) is a spacer-factored permutation test; the foundation models being probed are DNA models, not LLMs in the natural-language sense.